	Academic Semester	Fall 2026-2027	**Academic level**	3
	Course Code	MT301	**Exam**	Quiz 1 — Model D
	Course Title	Probability and Statistics I		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Mai Farag , Dr. Walid Abd-Elraheem		

Student Name: _____ **Student ID:** _____

*Answer **all** questions. Calculators are allowed. Show your work in Questions 2 and 3; give probabilities as simplified fractions.*

Question 1 — Choose the correct answer (4 points)

One point each: ½ for the correct letter, ½ for a correct reason. Circle the letter and write why you chose it in the box.

1. A restaurant offers 4 starters, 5 main dishes and 2 desserts. How many meals with one choice from each category are possible?

A) 11 **B)** 20 **C)** 42 **D)** 40

Why? (reason or calculation)

2. A club has 7 members. In how many ways can it appoint a president, a vice-president and a secretary, with a different person in each post?

A) 343 **B)** 21 **C)** 210 **D)** 35

Why? (reason or calculation)

3. Two fair dice are rolled. The probability that the sum of their faces is 10 is

A) $\frac{5}{36}$ **B)** $\frac{1}{12}$ **C)** $\frac{1}{36}$ **D)** $\frac{1}{9}$

Why? (reason or calculation)

4. A fair coin is tossed four times. The probability of getting **at least three heads** is

A) $\frac{5}{16}$ **B)** $\frac{1}{16}$ **C)** $\frac{1}{4}$ **D)** $\frac{3}{8}$

Why? (reason or calculation)

Question 2 — Counting (3 points)

A committee of 4 people is to be chosen from 5 men and 6 women.

(a) In how many ways can the committee be chosen? [1.5 pt]

(b) In how many ways can it be chosen if it must contain **exactly 3 women**? [1.5 pt]

Question 3 — Probability (3 points)

A 3-digit number is formed at random from the digits $\{1, 2, 3, 4, 5, 6, 7, 8\}$, with **no digit repeated**.

(a) Find the probability that the number is **even**. [1.5 pt]

(b) Find the probability that the number is **greater than 600**. [1.5 pt]

— End of quiz · Good luck —